

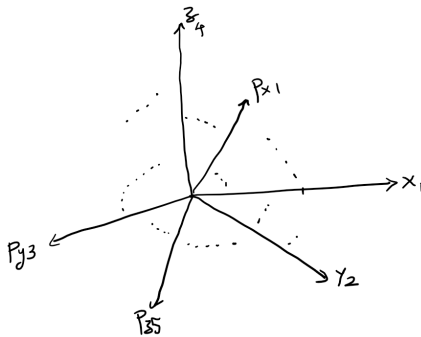
EP3120 - Statistical Mechanics

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Statistical mechanics of classical ideal gas - II

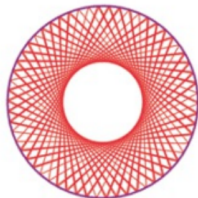
Phase space

- We studied the dynamics of the distribution function in the 6D phase space (that is also called as μ -space)
- Instead now we can imagine a $6N$ -dimensional space, representing the location and momentum of each particle
- In this space (called Γ -space) a particular state of the system is represented as a point
- As the system evolves with time, this point will move around in this $6N$ dimensional space

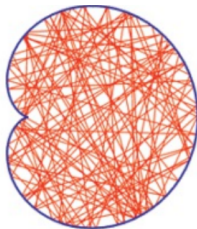


Ergodic hypothesis

- Depending on various constraints, the system is allowed to be in some region/s of the Γ -space
- The ergodic hypothesis states that - The system will cover the entire region of allowed Γ -space in a long enough time
- This means that time average can be thought of as an ensemble average



(a)



(b)

- The ergodic hypothesis can be proved mathematically for dynamical systems which satisfy certain properties
- However, it is not possible to prove it for all physical systems. Nevertheless it is believed to be true

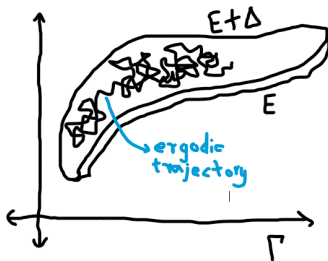
Ensembles

- The ergodic hypothesis implies that taking a long time statistical average of a quantity would be equivalent to taking an average over the possible different configurations that the system can take in Γ -space
- A collection of all the different possible configurations a system can take in the Γ -space is called an ensemble
- It is an imaginary construct of constructing multiple copies of the system with similar macro-properties (like volume, energy, pressure, temperature, etc.) defined as a “macro-state”
- These copies differ from each other in terms of their location in Γ -space, each location specifies a “micro-state”
- Instead of defining a distribution function f in μ -space, we can now describe the statistical properties of the system by defining a distribution function $\rho(\mathbf{r}_1, \dots, \mathbf{r}_N, \mathbf{p}_1, \dots, \mathbf{p}_N)$ such that the probability of finding the system in the vicinity of a particular microstate is

$$\rho(\mathbf{r}_1, \dots, \mathbf{r}_N, \mathbf{p}_1, \dots, \mathbf{p}_N) d\mathbf{r}_1 \dots d\mathbf{r}_N d\mathbf{p}_1 \dots d\mathbf{p}_N \quad (1)$$

Statistical postulate and microcanonical ensemble

- Consider a system that has energy E that is almost constant (within some error of measurement Δ)
- A constant energy surface would be a hyper-surface in Γ -space
- The trajectory of the system in Γ -space would be confined to a hyper-shell around this hyper-surface



- The ensemble of all the microstates that lie in this hyper-shell is called a micro-canonical ensemble
- The statistical postulate states that all micro-states inside the energy hypershell are equally probable

Statistical postulate

- The statistical postulate implies that

$$\rho = \begin{cases} \text{const}, & \text{if } E < \epsilon < E + \Delta \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

- The value of this constant ρ can be obtained by the condition that

$$\int d^N \mathbf{p} d^N \mathbf{r} \rho = 1 \quad (3)$$

$$\implies \text{const.} = \frac{1}{V_{\text{hypershell}}} \quad (4)$$

- Where the volume of the hyper-shell is (by definition)

$$V_{\text{hypershell}} = \int_{\text{hypershell}} d^N \mathbf{p} d^N \mathbf{r}$$

Entropy

- Similar to μ -space, ρ will also obey an equation of motion, similar to the Boltzmann transport equation
- The collisions will result in sudden jumps of the location of the system in Γ -space
- Inspired by the fact that Boltzmann's H is a quantity that never increases, while the entropy in an isolated system never decreases, we can relate

$$S = S_0 - K \int d^N \mathbf{p} d^N \mathbf{r} \rho \ln(\rho) \quad (5)$$

- Where S_0 and K are constants
- This integral will be non-zero only over the energy hypershell, in which ρ is a constant (outside of that $\rho \ln \rho \rightarrow 0$)
- Denote the volume of the hypershell as $\Gamma(E)$ and so ρ inside the hypershell becomes

$$\rho = \frac{1}{\Gamma(E)} \quad (6)$$

Coarse graining

- So the entropy becomes

$$S = S_0 - \frac{K}{\Gamma(E)} \ln \left(\frac{1}{\Gamma(E)} \right) \int d^N \mathbf{p} d^N \mathbf{r} = S_0 + K \ln[\Gamma(E)] \quad (7)$$

- Practically, we can measure a microstate with only limited precision
- There will be some error Δx in position measurement and Δp in momentum measurement
- That means we cannot distinguish between two microstates that lie within a volume - and so we count that as one microstate

$$(\Delta x)^{3N} (\Delta p)^{3N} \quad (8)$$

- This also has the benefit of making the number of microstates finite (which is otherwise infinite)
- Now we can define the number of microstates in the hyper-shell

$$\Omega(E) = \frac{\Gamma(E)}{\Delta x^{3N} \Delta p^{3N}} \quad (9)$$

Statistical definition of entropy

- Fixing some Δx and Δp and absorbing them in S_0 we can then write

$$S = S_0 + K \ln(\Omega) \quad (10)$$

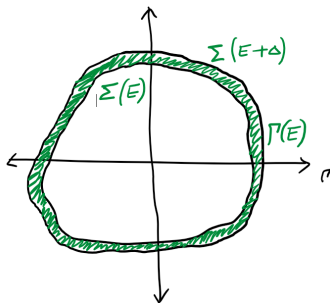
- Ω is dimensionless. In order to get the correct dimensions of entropy we can write

$$S = k_B \ln(\Omega) \quad (11)$$

- This is the Boltzmann formula for entropy (first written down by Planck who made the constant S_0 explicitly zero)
- This gives the absolute value of entropy
- It will clearly satisfy the third law of thermodynamics if there is only one microstate at absolute zero

Entropy of ideal gas

- Let's try to derive the entropy of an ideal gas from the Boltzmann formula
- Let's first calculate the volume of the hypershell $\Gamma(E)$
- Let $\Sigma(E)$ denote the hyper-volume in Γ -space which contains all microstates with energy $\epsilon \leq E$



- For $\Delta \ll E$, we can approximate the volume of the hyper-shell as

$$\Gamma(E) = \left(\frac{d\Sigma}{dE} \right) \Delta \quad (12)$$

Volume of hyper-sphere

- The definition of $\Sigma(E)$ is

$$\Sigma(E) = \int d\mathbf{r}_1 \dots d\mathbf{r}_N d\mathbf{p}_1 \dots d\mathbf{p}_N \quad (13)$$

$$= V^N \int d\mathbf{p}_1 \dots d\mathbf{p}_N \quad (14)$$

- The space part just integrates to the volume of the box for each particle
- The momentum integral is limited over that region of Γ -space that satisfies

$$\frac{p_{1x}^2}{2m} + \frac{p_{1y}^2}{2m} + \frac{p_{1z}^2}{2m} + \dots + \frac{p_{Nx}^2}{2m} + \frac{p_{Ny}^2}{2m} + \frac{p_{Nz}^2}{2m} \leq E \quad (15)$$

- This is the equation of a $3N$ dimensional sphere
- If the total momentum of gas particles is conserved, than that would impose additional constraints and the enegy hypershell would not be spherical, for ex., all particles moving in +ve x direction should not be an accessible microstate?

n -dimensional sphere

- Firstly, momentum is not conserved for a gas because of collisions with the box walls - all particles moving in the same direction is an allowed microstate
- Second - such microstates will occupy much smaller volume in the sphere compared to the vast majority of microstates which will have roughly net zero momentum
- Consider the volume of a n -dimensional sphere of radius R , Σ_{ns}

$$\Sigma_{ns} = \int dx_1 \dots dx_n \quad (16)$$

- with the condition $x_1^2 + x_2^2 + \dots x_n^2 \leq R^2$
- From the spherical symmetry of this volume and by dimensional analysis, we know that the volume should be of the form

$$\Sigma_{ns} = C_n R^n \quad (17)$$

n -dimensional sphere

- Consider the integral

$$I = \int_{-\infty}^{\infty} dx_1 \dots \int_{-\infty}^{\infty} dx_n e^{(-x_1^2 - x_2^2 - x_3^2 \dots - x_n^2)} \quad (18)$$

$$= \left(\int_{-\infty}^{\infty} dx e^{-x^2} \right)^n \quad (19)$$

$$= \pi^{n/2} \quad (20)$$

- Here we have used the $\gamma(1/2) = \sqrt{\pi}$ result
- This integral can also be evaluated in spherical coordinates by taking $x_1^2 + x_2^2 + \dots x_n^2 = R^2$
- Now this integral can also be written as an integral over concentric spherical shells of radius R going from zero to infinity

$$I = \int_{R=0}^{\infty} \Gamma(R) e^{-R^2} \quad (21)$$

- Exercise: Evaluate this

n -dimensional sphere

- We get

$$I = \int_{R=0}^{\infty} \Gamma(R) e^{-R^2} = \int_0^{\infty} \left(\frac{d\Sigma}{dR} \right) dR e^{-R^2} \quad (22)$$

$$= C_n n \int_0^{\infty} dR R^{n-1} e^{-R^2} \quad (23)$$

$$= C_n n \frac{1}{2} \gamma(n/2) \quad (24)$$

- Comparing the two expressions we get

$$C_n = \frac{\pi^{n/2}}{\frac{n}{2} \gamma(n/2)} = \frac{\pi^{n/2}}{\gamma(\frac{n}{2} + 1)} \quad (25)$$

- And hence the volume of a n dimensional sphere with radius R becomes

$$\Sigma_{ns} = \frac{\pi^{n/2}}{\gamma(\frac{n}{2} + 1)} R^N \quad (26)$$

Number of microstates

- Coming back to the ideal gas problem, the hyper-volume with energy less than E becomes

$$\Sigma(E) = V^N \frac{\pi^{3N/2}}{\gamma(\frac{3N}{2} + 1)} (2mE)^{3N/2} \quad (27)$$

- Now we get

$$\Gamma(E) = \frac{d\Sigma}{dE} \Delta = \Sigma \frac{3N}{2} \frac{\Delta}{E} \quad (28)$$

- Let us choose Δx and Δp by using the Heisenberg uncertainty principle which states that

$$\Delta x \Delta p \geq \frac{\hbar}{2} \quad (29)$$

- Lets take $\Delta x \Delta p = h$ and so the number of microstates becomes

$$\Omega = \frac{\Gamma(E)}{(\Delta x \Delta p)^{3N}} = \left(\frac{V}{h^3}\right)^N \frac{\pi^{3N/2}}{\gamma(\frac{3N}{2} + 1)} (2mE)^{3N/2} \left(\frac{3N\Delta}{2E}\right) \quad (30)$$

Entropy of ideal gas

- Making use of the Boltzmann formula for entropy, we get

$$S = k_B \ln(\Omega) = k_B \ln \left[\left(\frac{V}{h^3} \right)^N \frac{\pi^{3N/2}}{\gamma(\frac{3N}{2} + 1)} (2mE)^{3N/2} \left(\frac{3N\Delta}{2E} \right) \right] \quad (31)$$

$$= Nk_B \ln \left[\frac{V}{h^3} (2\pi mE)^{3/2} \right] - k_B \ln[\gamma(\frac{3N}{2} + 1)] + k_B \ln \left[\frac{3N\Delta}{2E} \right] \quad (32)$$

- The gamma function is an extension of the factorial function over non-integers

$$\gamma\left(\frac{3N}{2} + 1\right) = \left(\frac{3N}{2}\right)! \quad (33)$$

- The Stirling's approximation is

$$\ln(n!) \approx n \ln(n) - n \quad (34)$$

Entropy of ideal gas

- Using the Stirling approximation, we get

$$S = Nk_B \ln \left[\frac{V}{h^3} (2\pi mE)^{3/2} \right] - Nk_B \ln \left[\left(\frac{3N}{2} \right)^{3/2} \right] + \frac{3Nk_B}{2} + \quad (35)$$

$$k_B \ln \left[\frac{3N\Delta}{2E} \right] \quad (36)$$

- For the last term, the error in energy measurement is typically

$$\frac{\Delta}{E} \sim \frac{1}{\sqrt{N}} \implies k_B \ln \left[\frac{3N\Delta}{2E} \right] \sim k_B \ln[3N/2] \quad (37)$$

- This term is much smaller compared to the other terms which all are multiplied by N
- So this term can be ignored and it shows that the results do not depend on the error Δ

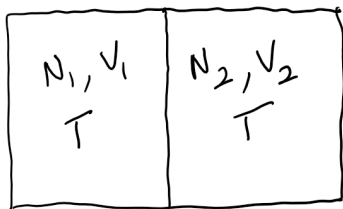
Entropy of ideal gas

- Finally the entropy expression becomes

$$S = Nk_B \ln \left[V \left(\frac{4\pi mE}{h^2 3N} \right)^{3/2} \right] + \frac{3}{2} Nk_B \quad (38)$$

- Question: Is this an extensive quantity?
- We can take $E \propto N$ in the thermodynamic limit, so E/N will be an intensive quantity
- Each term has a pre-factor N so they will scale as $\propto N$
- However, the V inside the $\ln$ will make the quantity not extensive
- This problem is stated explicitly in the Gibbs paradox

Gibbs paradox



- Consider an ideal gas with N_1 particles in volume V_1 and another ideal gas with N_2 particles and volume V_2 . Both have the same temperature T and pressure P
- The total entropy of this system will be

$$S = N_1 k_B \ln \left[V_1 \left(\frac{4\pi m E_1}{3h^2 N_1} \right)^{3/2} \right] + \frac{3}{2} (N_1 + N_2) k_B \quad (39)$$

$$+ N_2 k_B \ln \left[V_2 \left(\frac{4\pi m E_2}{3h^2 N_2} \right)^{3/2} \right] \quad (40)$$

Gibbs paradox

- If the partition is removed, then both gases will expand and fill out the volume $V_1 + V_2$
- Since the temperatures are same before and total energy is conserved, that means temperatures will remain same and energy of individual gases will also remain same
- After the partition is removed, the change in entropy becomes
- The change in entropy is

$$\Delta S = N_1 k_B \ln \left[\frac{V_1 + V_2}{V_1} \right] + N_2 k_B \ln \left[\frac{V_1 + V_2}{V_2} \right] \quad (41)$$

- However, if both the gases were identical to begin with, then removing the partition should not cause any change in entropy because there is no difference in the physical setup with or without the partition
- This is the paradox posed by J. W. Gibbs

Resolution of Gibbs paradox

- Gibbs resolved this paradox by stating that we have over-counted the number of microstates by a factor of $N!$
- So the correct number of microstates is actually $\Omega_{correct} = \Omega/N!$
- This will reduce the entropy by $S_{correct} = k_B \ln(\Omega') = S - \ln(N!)$
- Then by making the use of the Stirling approximation, we get a new formula for entropy

$$S = Nk_B \ln \left[V \left(\frac{4\pi mE}{h^2 3N} \right)^{3/2} \right] + \frac{3}{2} Nk_B - Nk_B \ln(N) + Nk_B \quad (42)$$

$$= Nk_B \ln \left[\frac{V}{N} \left(\frac{4\pi mE}{h^2 3N} \right)^{3/2} \right] + \frac{5}{2} Nk_B \quad (43)$$

- Recalculating the change in entropy now gives

$$\Delta S = (N_1 + N_2)k_B \ln \left[\frac{V_1 + V_2}{N_1 + N_2} \right] - N_1 k_B \ln \left[\frac{V_1}{N_1} \right] - N_2 k_B \ln \left[\frac{V_2}{N_2} \right] \quad (44)$$

- Here we have used the fact $E_1/N_1 = E_2/N_2 = (E_1 + E_2)/(N_1 + N_2)$

Resolution of Gibbs paradox

- Since the gases have the same pressure also, it means that
$$\frac{V_1}{N_1} = \frac{V_2}{N_2} = \frac{V_1+V_2}{N_1+N_2}$$
- This makes the change in entropy $\Delta S = 0$ and resolves the Gibbs paradox
- We can also now see the extensive nature of entropy explicitly from the expression
- The reduction of the number of microstates by the factor of $N!$ was simply postulated (much before advent of quantum mechanics)
- However, with the advent of quantum mechanics and the uncertainty principle, it became clear that at the quantum level, particles are indistinguishable
- This means that the $N!$ microstates where particles are at the same location in phase space, but just the identity of particles are interchanged - all these microstates are the same microstate and should be counted as one
- This is a striking example of quantum mechanics becoming necessary for classical system

Derivation of thermodynamics

- The entropy is expressed as a function $S(N, V, E)$. It can be inverted to express the energy E

$$E = \frac{3Nh^2}{4\pi m} \left(\frac{N}{V} \right)^{2/3} \exp \left[\frac{2}{3} \frac{S}{Nk_B} - \frac{5}{3} \right] \quad (45)$$

- Using the Maxwell relations we have

$$T = \left. \frac{\partial E}{\partial S} \right|_V = \frac{2}{3} \frac{E}{Nk_B} \quad (46)$$

- This recovers the thermodynamic result $E = (3/2)Nk_B T$
- We can also get the correct equation of state

$$P = - \left. \frac{\partial E}{\partial V} \right|_S = \frac{2E}{3V} = \frac{Nk_B T}{V} \quad (47)$$

Sackur Tetrode equation

- The equation derived for entropy

$$Nk_B \ln \left[\frac{V}{N} \left(\frac{4\pi m E}{h^2 3N} \right)^{3/2} \right] + \frac{5}{2} Nk_B \quad (48)$$

- is known as the Sackur-Tetrode equation
- As $T \rightarrow 0$, quantum mechanical effects will dominate
- In quantum mechanics, the particles have a zero-point energy (like for particle in a box). Taking it roughly as

$$\frac{E}{N} \approx \frac{h^2}{8\pi m} \left(\frac{N}{V} \right)^{2/3} \quad (49)$$

$$S \approx Nk_B \left[\frac{5}{2} - \frac{3}{2} \ln(6) \right] \quad (50)$$

- This can be close to zero - as required by third law of thermodynamics
- A more accurate treatment of the quantum mechanical ground state energy is needed to get the exact entropy at low temperatures